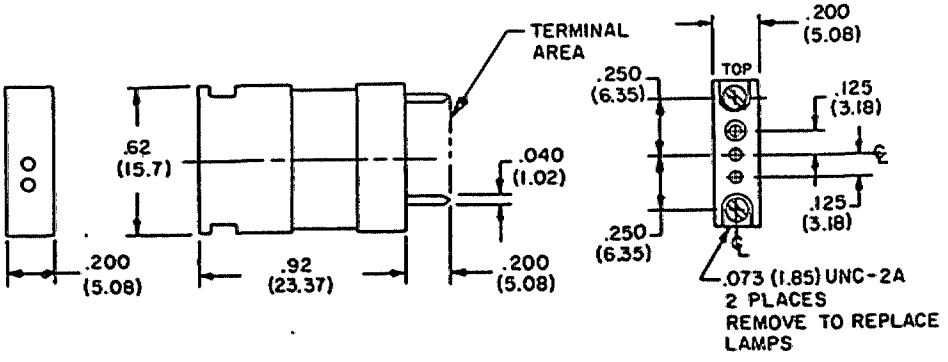
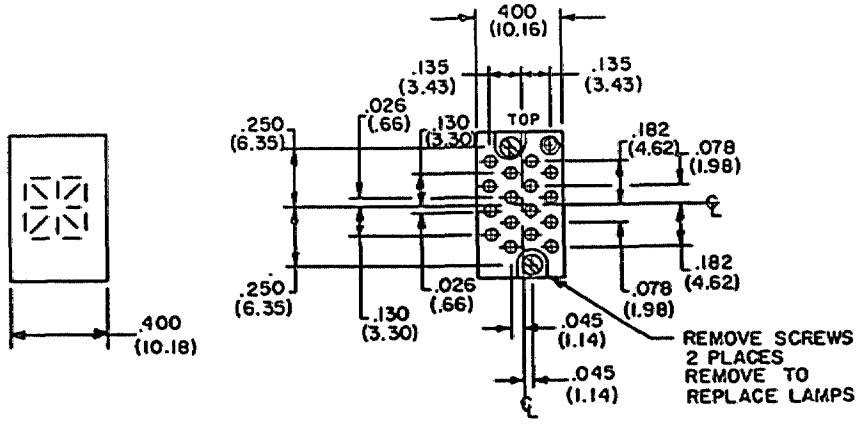
MILITARY SPECIFICATION SHEET

© DISPLAY, OPTOELECTRONIC, INDICATOR, DIGITAL, INCANDESCENT (STYLE I), SEGMENTED, DOT PATTERN AND BAR PATTERN, TYPE R04 (REPLACEMENT PARTS FOR TYPE R03 DISPLAY ASSEMBLY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

© The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-D-28803.



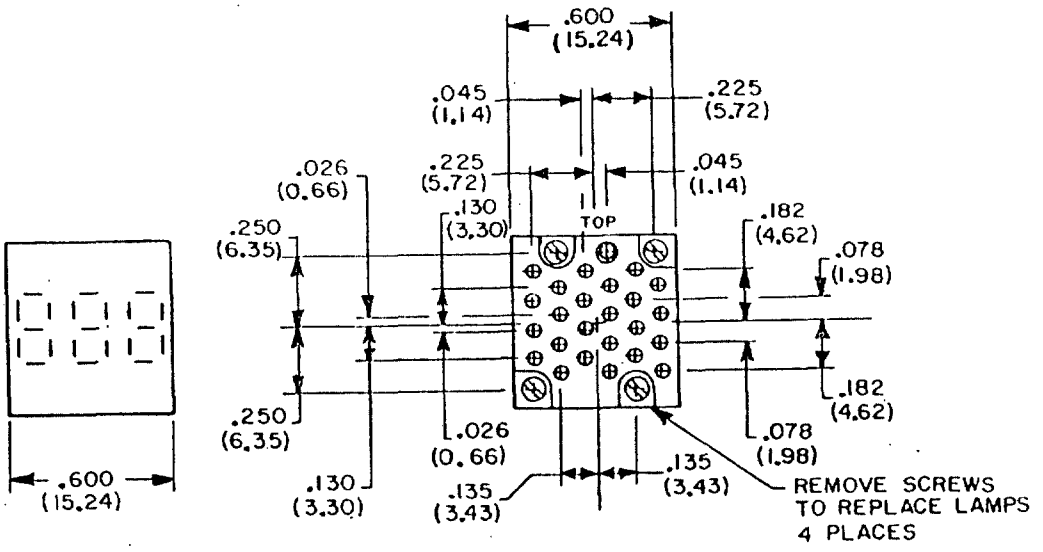


© FIGURE 1. Type R04 display module assemblies.

© denotes changes

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NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents are given for general information only.
4. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.
5. Design configuration optional with envelope as shown.
6. For segment dimensions, see figures 4, 7, and 9.

© FIGURE 1. Type R04 display module assemblies - Continued.

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TRUTH TABLE				
CIRCUIT CONFIG 1				
INPUT		DISPLAY	DOT	
A	B		A	B
○	○	COLON	△	△
	○	DECIMAL		△
	○	DEGREE		△

FIGURE 2. Truth table -2 dot direct wire colon display.

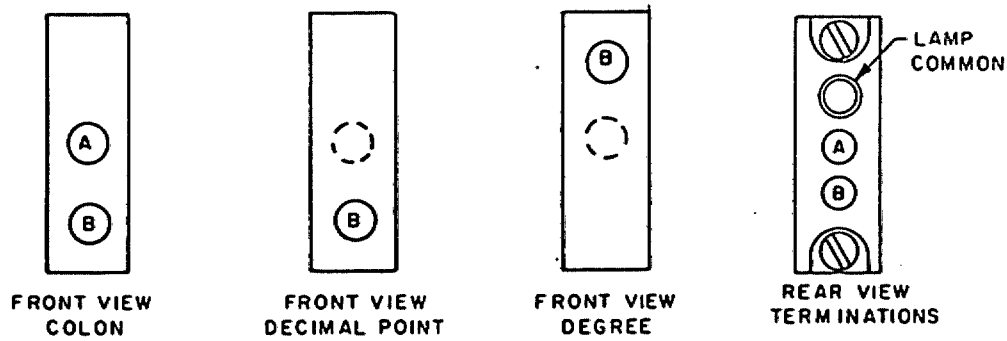


FIGURE 3. All colon display configurations.

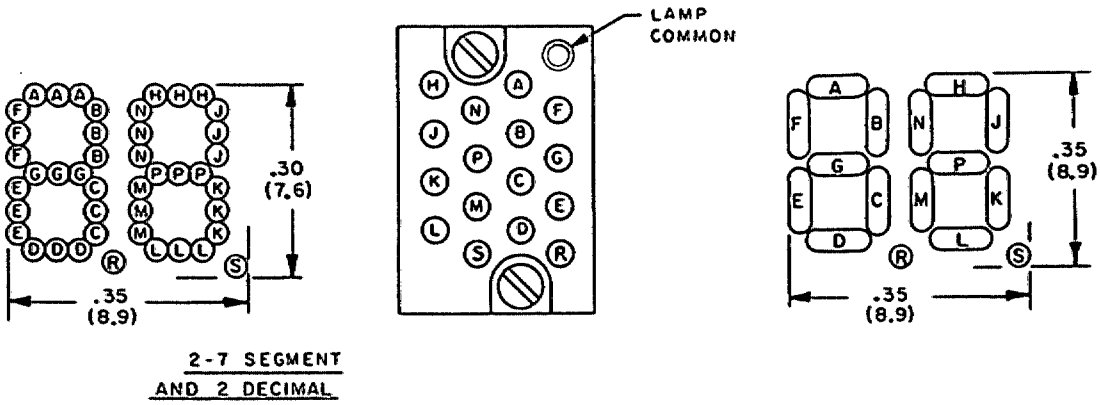


FIGURE 4. Segment dimensions and terminations for direct wired numeric alpha display.

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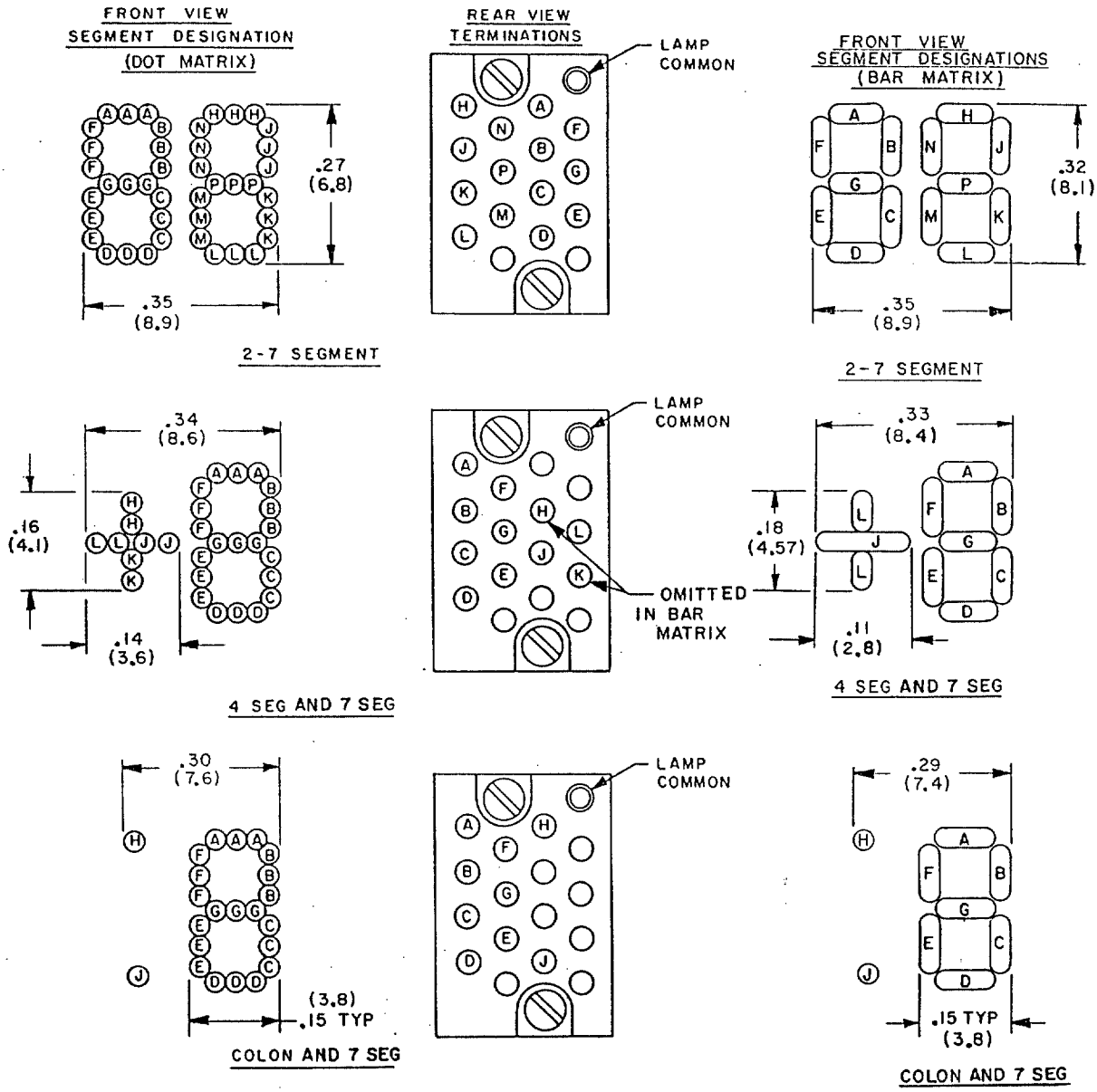


FIGURE 4. Segment dimensions and terminations for direct wired numeric alpha display - Continued.

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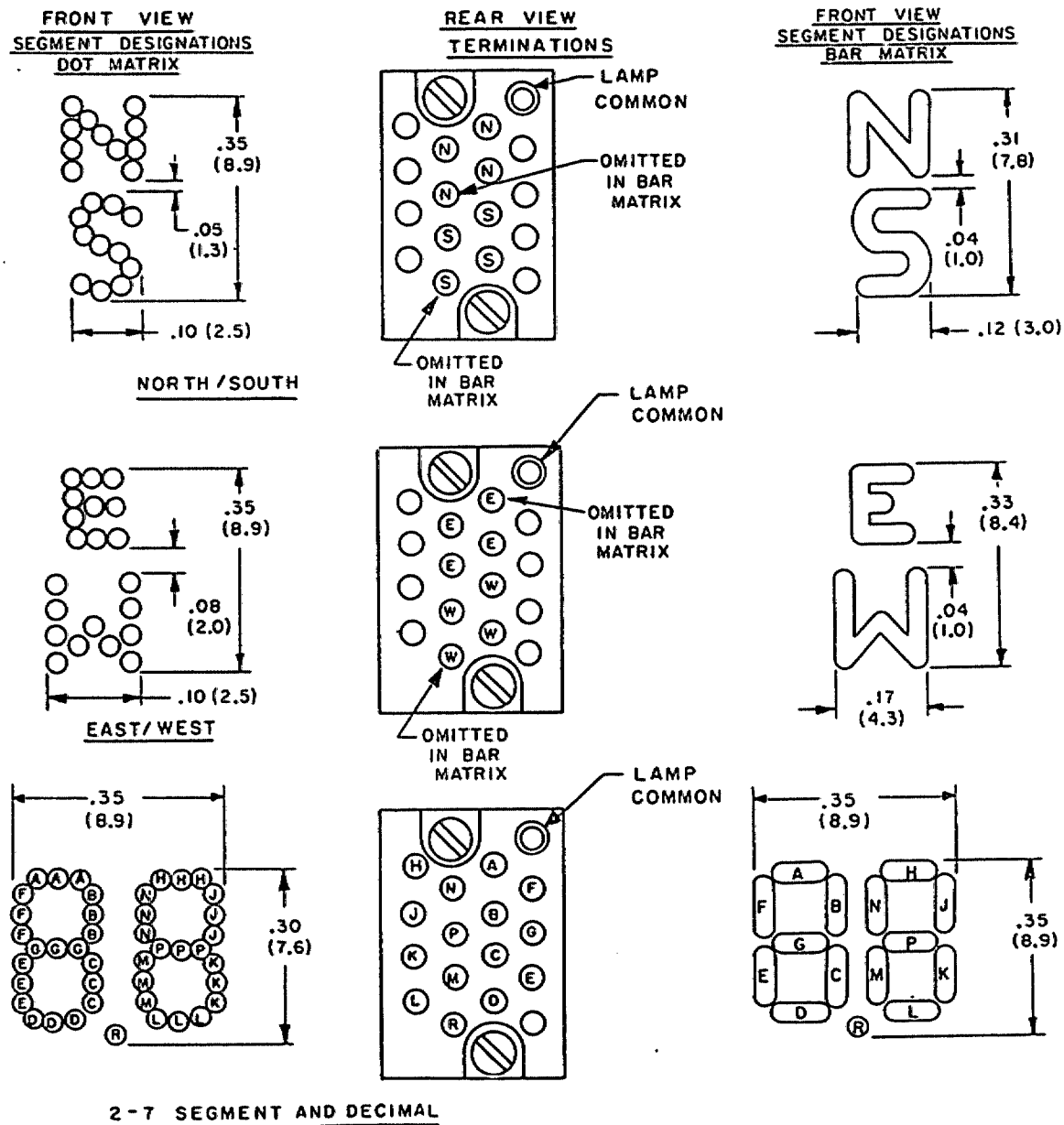


FIGURE 4. Segment dimensions and terminations for direct wired numeric alpha display - Continued.

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Check the source to verify that this is the current version before use.

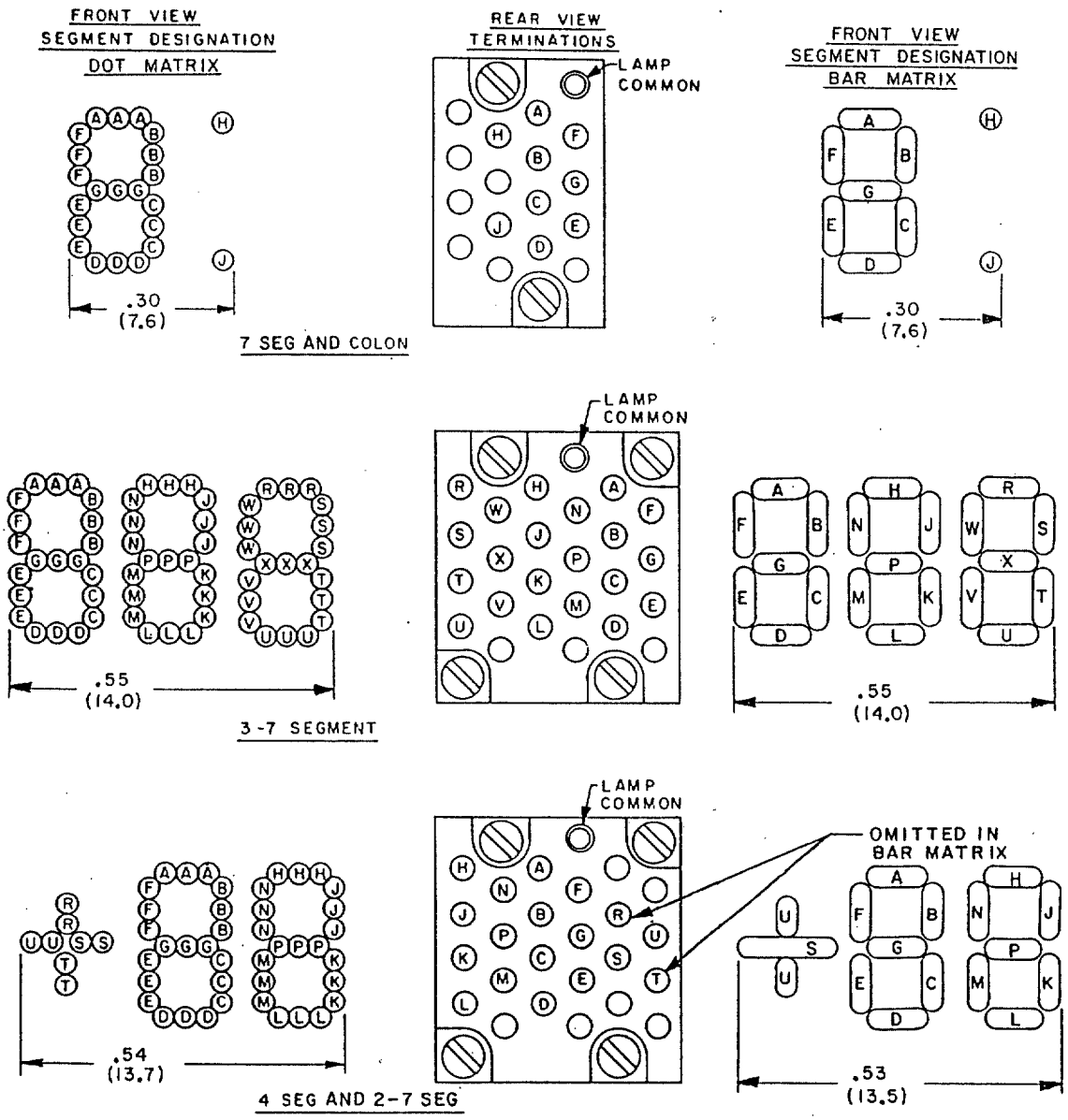
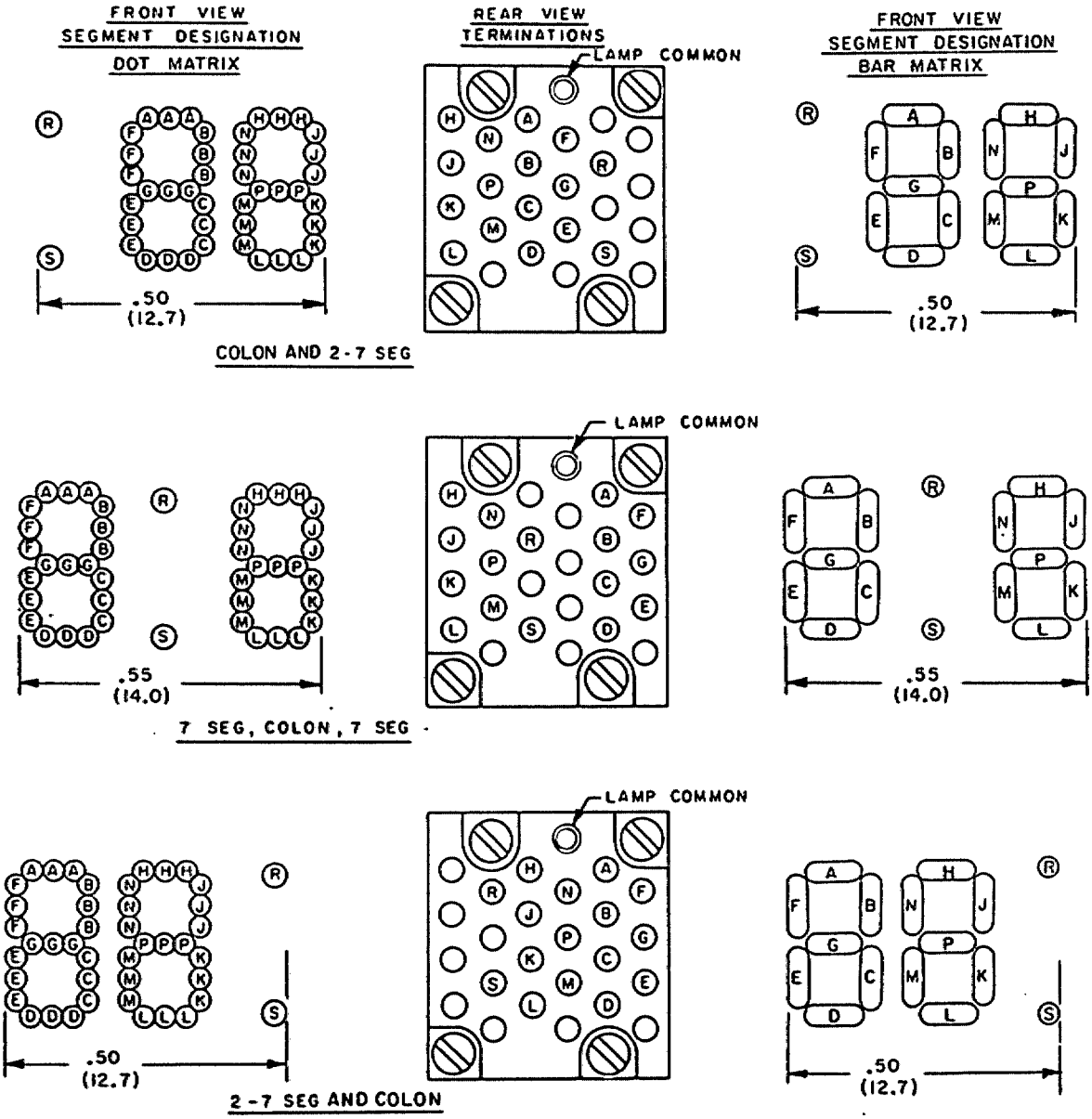


FIGURE 4. Segment dimensions and terminations for direct wired numeric alpha display - Continued.

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NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents are given for general information only.
4. Unless otherwise specified, tolerances are ±.02 (0.5 mm).

FIGURE 4. Segment dimensions and terminations for direct wired numeric alpha display - Continued.

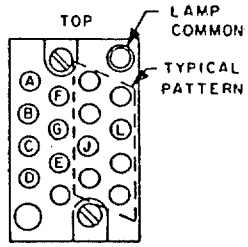
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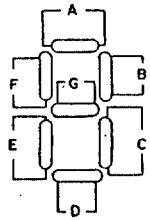
FRONT VIEW
BAR MATRIX

TRUTH TABLE								
CIRCUIT CONFIGURATION 2								
				DISPLAY				
J	L				J	L		
○	○			+	△	△		
○				−	△			

NOTE: Typical designators H through J = R through U.



REAR VIEW
TERMINATIONS



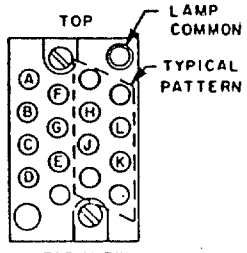
FRONT VIEW - 7 SEGMENT
BAR MATRIX



FRONT VIEW
DOT MATRIX

TRUTH TABLE								
CIRCUIT CONFIGURATION 2								
				DISPLAY				
H	J	K	L		H	J	K	L
○	○	○	○	+	△	△	△	△
	○		○	−		△		△

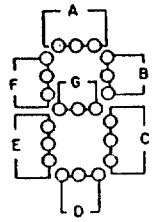
NOTE: Typical designators J and L = S and U.



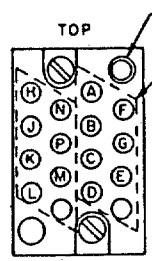
REAR VIEW
TERMINATIONS

FIGURE 5. Truth table for typical four segment direct wire sign display.

TRUTH TABLE														
CIRCUIT CONFIGURATION 3														
INPUT 2/							DISPLAY	SEGMENT LIT 2/						
A	B	C	D	E	F	G		A	B	C	D	E	F	G
○	○	○	○	○	○		0	△	△	△	△	△	△	
	○	○					1		△	△				
○	○		○	○		○	2	△	△		△	△		△
○	○	○	○			○	3	△	△	△	△			△
	○	○			○	○	4		△	△			△	△
○		○	○		○	○	5	△		△	△		△	△
○		○	○	○	○	○	6	△		△	△	△	△	△
○	○	○					7	△	△	△				
○	○	○	○	○	○	○	8	△	△	△	△	△	△	△
○	○	○	○		○	○	9	△	△	△	△		△	△
○	○	○		○	○	○	A	△	△	△		△	△	△
		○	○	○	○	○	b			△	△	△	△	△
○			○	○	○		C	△			△	△	△	
	○	○	○	○		○	d		△	△	△	△		△
○			○	○	○	○	E	△			△	△	△	△
○				○	○	○	F	△				△	△	△



FRONT VIEW - 7 SEGMENT
DOT MATRIX



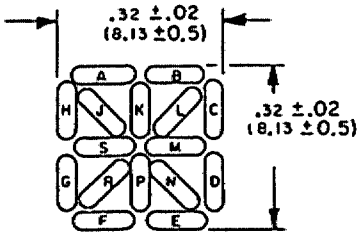
REAR VIEW - TERMINATION

NOTE: Typical 7 segment pattern repeats as A through G, H through P, and R through X (see figure 4).

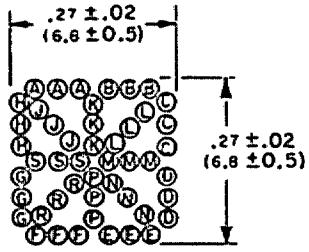
FIGURE 6. Truth table: typical seven segment direct wired numeric alpha display.

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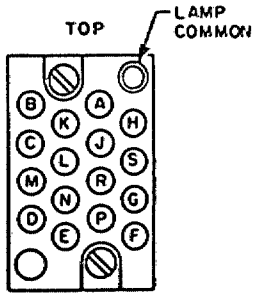
TRUTH TABLE CIRCUIT CONFIGURATION 4 INPUT / SEGMENT LIT																
DISPLAY	A	B	C	D	E	F	G	H	J	K	L	M	N	P	R	S
A	△	△	△	△			△	△				△				△
B	△	△	△	△	△	△				△		△		△		
C	△	△			△	△	△	△								
D	△	△	△	△	△	△				△				△		
E	△	△			△	△	△	△				△				△
F	△	△					△	△				△				△
G	△	△		△	△	△	△	△				△				
H			△	△			△	△				△				△
I	△	△			△	△				△				△		
J			△	△	△	△	△									
K							△	△			△		△			△
L					△	△	△	△								
M			△	△			△	△	△		△					
N			△	△			△	△	△				△			
O	△	△	△	△	△	△	△	△								
P	△	△	△				△	△				△				△
Q	△	△	△	△	△	△	△	△					△			
R	△	△	△				△	△				△	△			△
S	△	△		△	△	△		△				△				△
T	△	△								△				△		
U			△	△	△	△	△	△								
V							△	△			△				△	
W			△	△			△	△					△		△	
X									△		△		△		△	
Y									△		△			△		
Z	△	△			△	△					△				△	



FRONT VIEW
16 SEGMENT DESIGNATIONS
BAR MATRIX



FRONT VIEW
16 SEGMENT DESIGNATIONS
DOT MATRIX

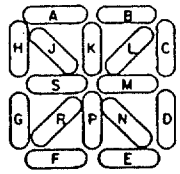


REAR VIEW
TERMINATIONS

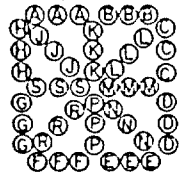
FIGURE 7. Truth table and segment dimensions: Alpha and numeric direct wire - alphabetic characters.

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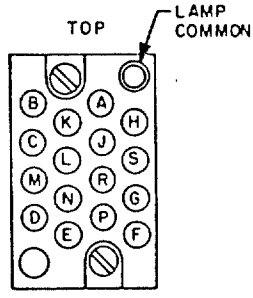
TRUTH TABLE CIRCUIT CONFIGURATION 5 INPUT / SEGMENT LIT																
DISPLAY	A	B	C	D	E	F	G	H	J	K	L	M	N	P	R	S
0		△	△	△	△					△				△		
1										△				△		
2		△	△		△							△		△		
3		△	△	△	△							△				
4			△	△						△		△				
5		△		△	△					△		△				
6		△		△	△					△		△		△		
7		△	△	△												
8		△	△	△	△					△		△		△		
9		△	△	△	△					△		△				
10		△	△	△	△		△	△		△				△		
11			△	△			△	△								
12		△	△		△		△	△				△		△		
13		△	△	△	△		△	△				△				
14			△	△			△	△		△		△				
15		△		△	△		△	△		△		△				
16		△		△	△		△	△		△		△		△		
17		△	△	△			△	△								
18		△	△	△	△		△	△		△		△		△		
19		△	△	△	△		△	△		△		△				
+										△		△		△		△
−												△				△
/											△				△	
\									△				△			
⅝	△	△		△	△	△		△		△		△		△		△
▷							△	△	△						△	
◁			△	△							△		△			



FRONT VIEW
16 SEGMENT DESIGNATIONS
BAR MATRIX



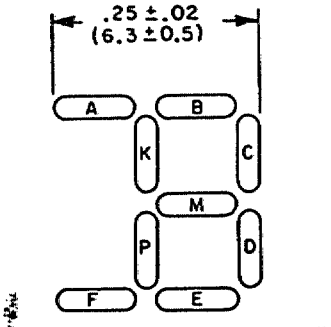
FRONT VIEW
16 SEGMENT DESIGNATIONS
DOT MATRIX



REAR VIEW
TERMINATIONS

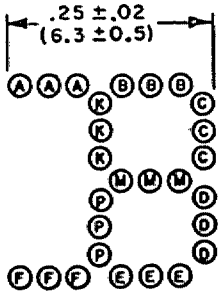
FIGURE 8. Truth table: Alpha and numeric direct wire - numeric characters and symbols.

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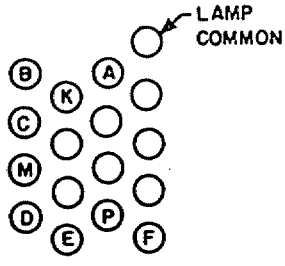


FRONT VIEW - 9 SEGMENT
BAR MATRIX

TRUTH TABLE CIRCUIT CONFIGURATION 6 INPUT/SEGMENT LIT									
	A	B	C	D	E	F	K	M	P
0		△	△	△	△		△		△
1			△	△					
2		△	△		△			△	△
3		△	△	△	△			△	
4			△	△			△	△	
5		△		△	△		△	△	
6				△	△		△	△	△
7		△	△	△					
8		△	△	△	△		△	△	△
9		△	△	△			△	△	
A		△	△	△			△	△	△
B	△	△	△	△	△	△	△	△	△
C		△			△		△		△
D	△	△	△	△	△	△	△		△
E		△			△		△	△	△
F		△					△	△	△



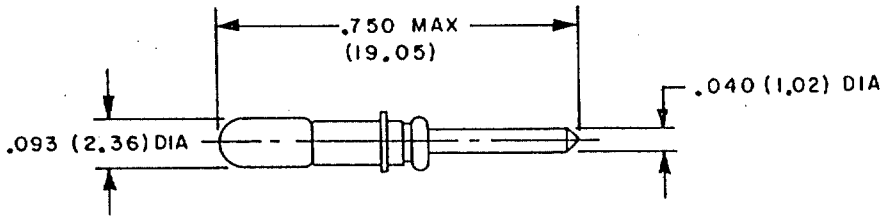
FRONT VIEW-9 SEGMENT
DOT MATRIX



REAR VIEW TERMINATION

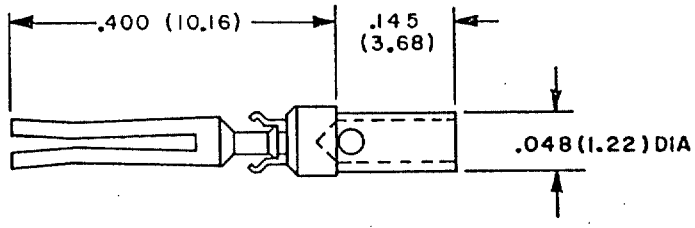
FIGURE 9. Truth table and segment dimensions: Nine segment direct wired numeric alpha display.

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NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents are given for general information only.
4. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
5. Design configuration optional with envelope as shown.

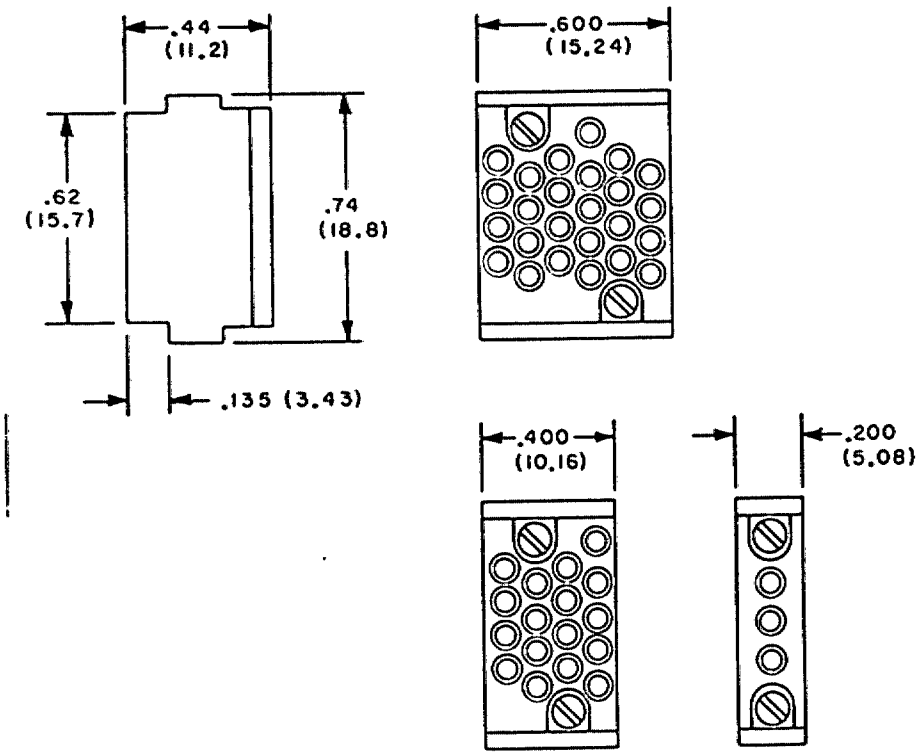
FIGURE 10. Lamp assembly.



NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents are given for general information only.
4. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
5. Design configuration optional with envelope as shown.

© FIGURE 11. Terminal, crimp style.

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NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents are given for general information only.
4. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.
5. Design configuration optional with envelope as shown.

FIGURE 12. Connector blocks.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:59Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 through figure 12 and table I through table II.

Weight: See table I.

Light source: T-3/4 incandescent.

Storage temperature: -55°C to +85°C.

© Operating temperature: -55°C to +71°C.

Vibration grade: 2 installed.

Shock: Method I installed.

Chromaticity: Method I.

Luminance: Average 300-foot-lamberts minimum with a supply voltage of 5 volts.

Dot-to-dot ratio:

Segment to segment: 3:1 maximum.

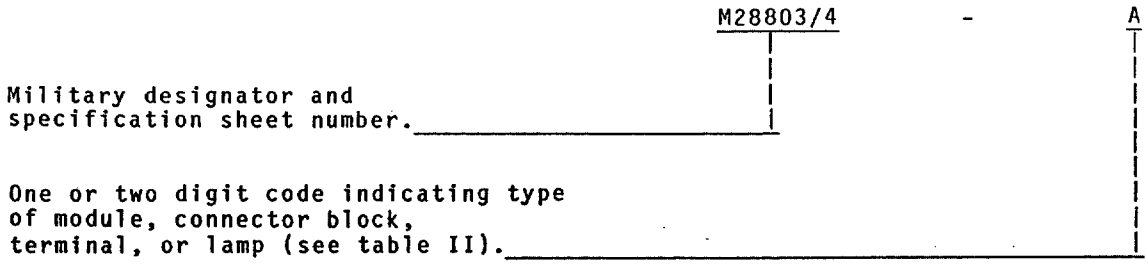
Digit to digit: 3:1 maximum.

Electrical:

Supply voltage: 5 volts maximum.

Supply current: (Each lamp) .025 ampere maximum.

Part or Identifying Number (PIN): The PIN shall consist of the letter "M" followed by the basic number of the specification sheet and an one or two digit symbol indicating the type of module, connector block, terminal, or lamp.



© Intended use: MIL-D-28803/4 is intended for use as a part of display assemblies covered by MIL-D-28803/3.

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TABLE I. Weight of modules (by code letter). 1/

Module code letter	Maximum weight in (grams)	Module code letter	Maximum weight in (grams)
A-A1	6.0	L-L1	11.0
B-B1	8.0	M-M1	8.0
C-C1	8.0	N	2.0
D-D1	8.0	P	6.0
E-E1	8.0	Q	8.0
F-F1	8.0	R	2.0
G-G1	11.0	S	0.2
H-H1	11.0	T-U-T1-U1	8.0
J-J1	11.0	V-V1	8.0
K-K1	11.0	W-W1	8.0

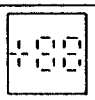
1/ For explanation of module code letter, see table II.

TABLE II. Code letters for type of module assemblies connector blocks, terminals and lamp assemblies.

Bar matrix Code letter	Dot matrix Code letter	Display	Type of module
A1	A		Colon, degree or decimal
B1	B		16 segment alpha/numeric
C1	C		2 - 7 segment numeric
D1	D		4 segment sign and 7 segment numeric
E1	E		Colon and 7 segment numeric
F1	F		7 segment numeric and colon
G1	G		3 - 7 segment numeric

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TABLE II. Code letters for type of module assemblies connector blocks, terminals and lamp assembies - Continued.

Bar matrix Code letter	Dot matrix Code letter	Display	Type of module
H1	H		1 - 4 segment and 2 - 7 segment numeric
J1	J		Colon and 2 - 7 segment numeric
K1	K		7 segment, colon and 7 segment numeric
L1	L		2 - 7 segment numeric and colon
M1	M		1 - 9 segment alpha numeric
N	N	.200 wide	connector block (figure 6) 1/
P	P	.400 wide	connector block (figure 6) 1/
Q	Q	.600 wide	connector block (figure 6) 1/
R	R	---	Terminals (figure 5) 10 to the bag
S	S	---	Lamp assembly (figure 4)
T1	T		North-South
U1	U		East-West
V1	V		2 - 7 segment with decimal
W1	W		2 - 7 segment with 2 decimal

1/ For Government acquisitions, connector blocks shall be furnished with the required number of terminals plus two extra.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - AS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR, AT, AV, ME, MI
Navy - MC, OS, SH

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5980-0004-4)

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